XING HAN

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Research Interests

I build multimodal foundation models and agents that acquire the core competencies of human experts: integrating heterogeneous evidence as it unfolds over time, reasoning about "what-if" trajectories, and staying reliable as the situation evolves. My work bridges core machine learning methodology with high-stakes applications such as medicine.

- **Multimodal Foundation Models:** multimodal fusion, cross-modal interactions, clinical foundation models
- **Counterfactual Reasoning & Simulation:** counterfactual simulation, physiology-constrained generative models
- **Trustworthy & Enduring Deployment:** uncertainty quantification, sequential monitoring, continual learning
- **Foundations of Machine Learning:** neural scaling laws, time-series sequence models, interpretability

Education

M.Sc - Ph.D. in Electrical and Computer Engineering — August 2023
The University of Texas at Austin — Austin, TX, USA
Advisor: Joydeep Ghosh Committee: Aryan Mokhtari, Atlas Wang, Mingyuan Zhou, Nhat Ho
Dissertation: Reliable Predictions for Structured and Corrupted Data

B.Eng. in Electrical and Electronics Engineering — July 2017
The University of Edinburgh — Edinburgh, UK
Advisor: Jiabin Jia GPA: 3.92 (graduated with first-class honor)
University of Edinburgh "2+2" Scholarship in 2015 & 2016

Publications

XH denotes the author; * denotes equal contribution; ✉ denotes corresponding author.

Selected Papers

[1] **On the Invariance and Generality of Neural Scaling Laws**
Top ~3% NeurIPS submissions by reviewer rating
XH, Z. Liu, S. Saria, and P. Liang
Preprint 2026

[2] **FLAME: Adaptive Mixture-of-Experts for Continual Multimodal Multi-Task Learning**
XH, S. Chaudhari, T. Ranade, R. Chellappa, and S. Saria
Preprint 2026

[3] **Massively Multimodal Foundation Models: A Frameworks for Capturing Interactions with Specialized Mixture-of-Experts**
Top ~10% of accepted papers by reviewer rating
XH, H. Chung, J. Ghosh, P. Liang, and S. Saria
International Conference on Learning Representations (ICLR) 2026

[4] **QoQ-Med3: Robust Multimodal Clinical Analysis Foundation Model with Reasoning**
W. Dai, J. She, J. Cheong, **XH**, C. Harris, H. Wei, F. Vahedifard, S. Saria, R. Stevens, and P. Liang
NPJ Digital Medicine 2026

[5] **WATCH: Adaptive Monitoring for AI Deployments via Weighted-Conformal Martingales**
D. Prinster ✉, **XH**✉, A. Liu, and S. Saria
International Conference on Machine Learning (ICML) 2025

[6] **FuseMoE: Mixture-of-Experts Transformers for Fleximodal Fusion**
Over 150 citations as of July 2026
XH, H. Nguyen, C. Harris, N. Ho, and S. Saria
Neural Information Processing Systems (NeurIPS) 2024

[7] **Designing Robust Transformers using Robust Kernel Density Estimation**
XH, T. Ren, T. Nguyen, K. Nguyen, J. Ghosh, and N. Ho
Neural Information Processing Systems (NeurIPS) 2023

[8] **Architecture Agnostic Federated Learning for Neural Networks**
D. Makhija, **XH**, N. Ho, and J. Ghosh
International Conference on Machine Learning (ICML) 2022

[9] **Simultaneously Reconciled Quantile Forecasting of Hierarchically Related Time Series**
XH, S. Dasgupta, and J. Ghosh
Artificial Intelligence and Statistics (AISTATS) 2021

[10] **Certified Monotonic Neural Networks**
Spotlight presentation: $280/9454 \approx 2.96\%$; over 185 citations as of July 2026
X. Liu, **XH**, N. Zhang, and Q. Liu
Neural Information Processing Systems (NeurIPS) 2020

Other Papers

[11] **MILM: Large Language Models for Multimodal Irregular Time Series with Informative Sampling**
H. Chung, S. Li, Y. Wald, **XH**, S. Saria, and J. Ghosh
Preprint 2026

[12] **EvoMed-Agent: Self-Evolving Reasoners for Multimodal Clinical Sequences via Active Counterfactual Verification**
XH, Y. Wang, C. Chen, H. Chung, S. Li, G. Gudur, W. Dai, P. Liang, and S. Saria
In submission 2026

[13] **MINT: Multimodal Instruction Tuning with Multimodal Interaction Grouping**
X. Shan*, Q. Cao*, **XH***, H. Yu*, and P. Liang
Preprint 2025

[14] **Between Linear and Sinusoidal: Rethinking the Time Encoder in Dynamic Graph Learning**
H. Chung, S. Chaudhari, **XH**, Y. Wald, S. Saria, and J. Ghosh
Transactions on Machine Learning Research (TMLR) 2025

[15] **SSL-DA: Semi- and Self-Supervised Learning with Dual Attention for Echocardiogram Segmentation**
L. Lv, **XH**, Z. Sun, Z. Li, X. Wang, T. Jiang, Y. Liu, T. Li, J. Xu, L. You, G. Yao, F. Sun, and J. Xing
Journal of Imaging Informatics in Medicine 2025

[16] **Quadratic Gating Functions in Mixture of Experts: A Statistical Insight**

P. Akbarian*, H. Nguyen*, **XH***, and N. Ho
In submission to SIAM Journal on Mathematics of Data Science 2024

[17] **On Expert Estimation in Hierarchical Mixture of Experts: Beyond Softmax Gating Functions**
H. Nguyen*, **XH***, C. Harris, N. Ho, and S. Saria
In submission to JMLR 2024

[18] **Achieving Fairness Across Local and Global Models in Federated Learning**
D. Makhija, **XH**, J. Ghosh, and Y. Kim
Preprint 2024

[19] **Novel Node Category Detection Under Subpopulation Shift**
H. Chung, S. Chaudhari, Y. Wald, **XH**, and J. Ghosh
European Conference on Machine Learning and Data Mining (ECML-PKDD) 2024

[20] **Efficient Forecasting of Large Scale Hierarchical Time Series via Multilevel Clustering**
XH, T. Ren, J. Hu, J. Ghosh, and N. Ho
International Conference on Time Series and Forecasting (ITISE) 2023

[21] **A Novel Control-Variates Approach for Performative Gradient-Based Learners with Missing Data**
XH, J. Hu, and J. Ghosh
International Joint Conference on Neural Networks (IJCNN) 2023

[22] **Dynamic Combination of Heterogeneous Models for Hierarchical Time Series**
XH, J. Hu, and J. Ghosh
International Conference on Data Mining Workshops (ICDMW) 2022

[23] **Split Localized Conformal Prediction**
XH, Z. Tang, J. Ghosh, and Q. Liu
ICML Workshop on Distribution-Free Uncertainty Quantification 2021

[24] **Multi-Pair Text Style Transfer for Unbalanced Data via Task-Adaptive Meta-Learning**
XH and J. Lundin
ACL MetaNLP Workshop 2021

[25] **Model-Agnostic Explanations using Minimal Forcing Subsets**
XH and J. Ghosh
International Joint Conference on Neural Networks (IJCNN) 2021

[26] **Transparent Interpretation with Knockout**
XH, Y. Feng, N. Zhang, and Q. Liu
ICML Workshop on Human Interpretability in Machine Learning (WHI) [Spotlight] 2020

[27] **Sensing Personality to Predict Job Performance**
S. Lin, S.M. Mattingly, **XH**, P. Audia, et al.
CHI Workshop on The Future of Work 2019

[28] **Effects of Integrated Intent Recognition and Communication on Human-Robot Collaboration**
A. Gutierrez, M.L. Chang, **XH**, and K.C. Chang
International Conference on Intelligent Robots and Systems (IROS) 2018

Patents

[1] **Machine-Learning Based Generation of Text Style Variations for Digital Content Items**
J. Lundin, O. Schoppe, **XH**, M. Sollami, B. Lonsdorf, A. Ross, D. Woodward, and S. Rohde
US Patent 2022/0245322 A1 2022

Academic Employments

Postdoctoral Fellow, Johns Hopkins University — Baltimore, MD & Mountain View, CA
Supervisors: Prof. Suchi Saria, Prof. Gregory D. Hager — *09/01/2023 – now*

- Developing multi-modal Transformers and agents that integrate vital signs, lab results, clinical notes, and chest X-rays, enhancing predictive accuracy/reasoning of multimodal models with an emphasis on clinical applications

Graduate Research Assistant, The University of Texas at Austin — *06/01/2019 – 08/14/2023*
Intelligent Data Exploration and Analysis Lab, Department of Electrical and Computer Engineering
Supervisor: Prof. Joydeep Ghosh — Austin, TX

- Research topics include time-series forecasting, clustering, uncertainty quantification, federated learning, robustness and interpretability of machine learning

Data Science Group, Department of Statistics and Data Sciences
Supervisor: Prof. Nhat Ho — Austin, TX

- Research topics include optimal transport, Bayesian deep learning, model fusion/alignment, mixture of experts, and robustness of Transformers

Statistical Learning and AI Group, Department of Computer Science
Supervisor: Prof. Qiang Liu — Austin, TX

- Research topics include the robustness and interpretability of deep neural networks, conformal prediction, generative models, and predictive modeling of mobile sensor data

Industry Employments

Google — Sunnyvale, CA
Research Scientist Intern (Host: Dr. Xu Gao) — *05/31/2022 – 09/02/2022*

- Enhanced the workload prediction accuracy for GCP machines by 20% compared to baseline
- Utilized explanation tools (Shapley value) to understand the root causes of machine downtime

Intuit — Mountain View, CA
Data Scientist Intern (Mentor: Dr. Jing Hu) — *06/15/2021 – 09/03/2021*

- Improved cash-flow forecasting accuracy by 10% and deployed models to applications
- Decreased computation time by 45% through the development of a parallelized program

Salesforce — San Francisco, CA
Research Intern (Mentor: Dr. Jessica Lundin) — *06/01/2020 – 08/21/2020*

- Implemented 3 different state-of-the-art language models (LM) for text style transfer
- Leveraged meta-learning to fine-tune pre-trained LM on a small and unbalanced dataset

CognitiveScale — Austin, TX
Applied Scientist Intern (Mentor: Dr. Suyog Jain) — *05/29/2018 – 08/03/2018*

- Reproduced and improved 5 different session-based recommendation algorithms
- Led a team project of 4 interns to research “Ethical AI” and pitched a business idea to company

Grants & Protocols

Grant Drafter and Key Personnel, Intuit AI Research, **$300K** in total *2020 – 2023*

- Drafted grant proposals and budget planning documents for $\sim$ \$100,000/year from **Intuit AI Research** for three consecutive years. Developed a series of models in time-series forecasting and clustering, deployed in industry-level financial forecasting pipelines. Awarded annually to Dr. Joydeep Ghosh.

Core Contributor, Gordon and Betty Moore Foundation Grant, $\sim$ **$500K** in total *2023 – now*

- Developed multimodal clinical foundation models and post-deployment continuous monitoring methods that served as core technical contributions. Awarded annually to Dr. Suchi Saria.

Protocol Drafter and Submitter, Johns Hopkins Medicine IRB *2024 – now*

- Drafted, submitted, and maintained annual renewals for the following IRB protocols:
  - **IRB00464029**, "Leveraging Multimodal Patient Data for Enhanced Clinical Task Performance"
  - **CR00054024**, "Development and Validation of Robust Models for Adverse Event Risk Prediction"

Community Services

Reviewer, International Conference on Learning Representations (ICLR)	2020 – now
Reviewer, Neural Information Processing Systems (NeurIPS)	2020 – now
Reviewer, International Conference on Machine Learning (ICML)	2021 – now
Reviewer, Artificial Intelligence and Statistics (AISTATS)	2021 – now
Reviewer, International Joint Conference on Neural Networks (IJCNN)	2023 – now
Reviewer, Machine Learning for Health (ML4H)	2023 – now
Reviewer, International Conference on Acoustics, Speech and Signal Processing (ICASSP)	2023 – now
Reviewer, Machine Learning for Healthcare (MLHC)	2024 – now
Reviewer, SIAM Conference on Data Mining (SDM)	2026 – now
Journal Reviewer, Pattern Recognition	2019 – now
Journal Reviewer, Journal of Open Source Software	2024 – now
Journal Reviewer, IEEE Transactions on Pattern Analysis and Machine Intelligence (T-PAMI)	2025 – now
Journal Reviewer, Nature Biomedical Engineering	2025 – now
Journal Reviewer, Medical Image Analysis	2026 – now
Mentor, ML4H Author & Reviewer & Career Mentorship Program	2024 – now
Technical Program Committee, WellComp Workshop at UbiComp/ISWC	2026
Reading Group Organizer, Multimodal Learning in Health at JHU	2024 – 2025
Member, UT-Austin ECE Graduate Admission Committee	2021 – 2023
Volunteer, International Conference on Machine Learning (ICML) Workshops	2020, 2021, 2023
Third-year Class Representative, University of Edinburgh	2015 – 2016

Supervision

Shravan Chaudhari, CS Ph.D. Student	JHU, 2023 –
Carl Harris, Biomedical Engineering Ph.D. Student	JHU, 2023 –
Chen Chen, CS M.Sc. Student	JHU, 2026 –
Yuxin Wang, Applied Mathematics M.Sc. Student	JHU, 2026 –
Jiarui Shao, CS M.Sc. Student	JHU, 2026 –
Tanvi Ranade, CS B.Sc. + M.Sc. Student	JHU, 2024 – 2026
Hsing-Huan Chung, ECE Ph.D. Student	UT-Austin, 2023 – 2026 (now with Uber)
Disha Makhija, ECE Ph.D. Student	UT-Austin, 2023 – 2024 (now with Amazon)

Kurtis E. David, CS M.Sc. Student — UT-Austin, 2019 – 2020 (now with DeepMind)

Selected Honours and Awards

ICML Gold Reviewer Award	2026
NeurIPS Scholar Award, $2000	2023 – 2024
Excellent Teaching Assistant Award, $1000	2018
Outstanding Graduates of Beijing City (top 3%)	2017
University of Edinburgh 2 + 2 Scholarship, £2500 per year	2015 – 2016
Comprehensive Scholarship, ¥5000	2014 – 2015
China National Scholarship, ¥8000	2013 – 2014

Teaching Experience

Course Design and Trial Teaching, JHU Teaching Institute Certificate	2025
Teaching Assistant, CS 395T, Deep Probabilistic Modeling	UT-Austin, 2019
Teaching Assistant, ECE 422C, Software Design & Implementation	UT-Austin, 2018
Lab Instructor, ELEE09024, Digital System Design 3	University of Edinburgh, 2016

Invited Talks & Presentations

Guiding Mixture-of-Experts in Multimodal Learning: Temporal Dynamics and Robust Fusion	*EcoSta*, 2025
FuseMoE: Mixture-of-Experts Transformers for Fleximodal Fusion	*MIT Media Lab*, 2025
FuseMoE: Mixture-of-Experts Transformers for Fleximodal Fusion	*Columbia University*, 2024
Reliable Predictions for Structured and Corrupted Data	*Fudan University*, 2023
Hierarchical Time Series Forecasting	*IIF Workshop on Forecast Reconciliation*, 2023
Reliable Prediction of Treatment Effect Through Time	*Intuit AI*, 2023
Time Series Reconciliation with Dynamically Changing Structures	*Google Cloud AI*, 2022
Mixture-of-Experts for Probabilistic Forecasts of Aggregated Time Series	*Intuit AI*, 2021
Certified Monotonic Neural Networks	*Salesforce Research*, 2020
Transparent Interpretation with Knockout	*Intuit AI*, 2019

Technical Skills

Programming: Python (expert); R, Matlab (proficient); Java, Verilog, C/C++ (knowledge of)
Frameworks: PyTorch, TensorFlow, Scikit-Learn, Wandb, HuggingFace, TIMM, NLTK
Language: English (fluent), Mandarin (native), French (beginner)
Others: LaTeX, Shell Script, ROS, SQL

References

Suchi Saria, John C. Malone Associate Professor (Postdoc Advisor) — ssaria@cs.jhu.edu
Department of Computer Science, Johns Hopkins University — Baltimore, MD & New York, NY

Joydeep Ghosh, Schlumberger Centennial Chair Professor (PhD Advisor) — jghosh@utexas.edu
Department of Electrical and Computer Engineering, UT-Austin — Austin, TX

Paul Pu Liang, Assistant Professor — ppliang@mit.edu
Media Lab and EECS, MIT — Cambridge, MA